5.5 Solving Equations by Graphing

Name _____ Period ____ Date _____

Algebra I 5.5 · Student sheet

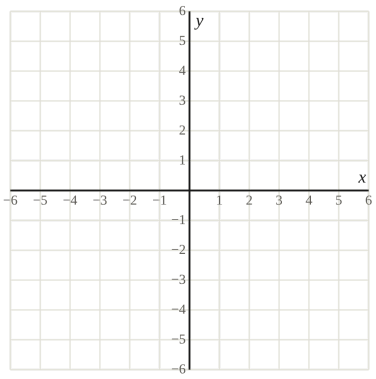
I can…

- solve linear equations by graphing.
- solve absolute value equations by graphing.
- use graphs to solve real-life problems.

Part 1 · Two sides, two graphs

Solve $-x + 1 = 2x - 5$ algebraically: _____

Now graph $y = -x + 1$ and $y = 2x - 5$. Intersection: _____



Graph $y = |x - 1|$ and $y = 2$ on the same grid. Solutions of $|x - 1| = 2$: _____

Part 2 · Notes: fill in the blanks

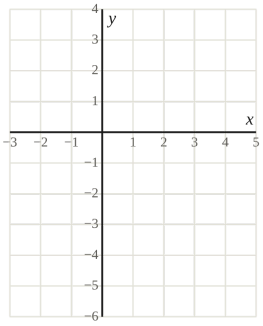
To solve $f(x) = g(x)$, graph $y =$ _____ and $y =$ _____.

The solution is the _____-coordinate of each intersection.

Part 3 · Examples (work them with the class)

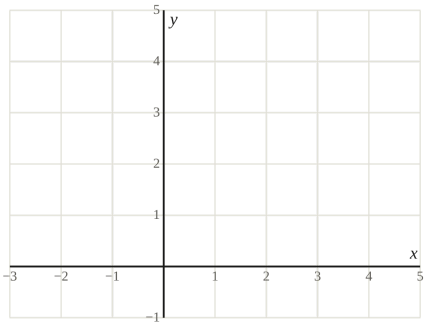
Example 1

Solve $-x + 1 = 2x - 5$ by graphing.



Example 2

Solve $|x - 1| = 2$ by graphing.



Example 3

You have \$40 and save \$5 a week. Your friend has \$10 and saves \$15 a week. Graph to find when you have the same amount.

Part 4 · Partner practice

Partner 1

Solve $\frac{1}{2}x + 2 = -x + 5$ by graphing.

Partner 2

Solve $|x + 2| = x + 4$ by graphing.